# NP - MIT

P = Problems that can be solved in polynomial time ($n^{O(1)}$) (size $n$)

NP = ~~Problem~~ {decision problems solvable in nondeterministic polynomial time}

answer is YES or NO

Can guess one out of polynomially many options in $O(1)$ time.

→ if any guess would lead to a YES, then we get such a guess.

Example:

3SAT: Given Boolean formula of the form

$$(x_1 \vee x_3 \vee \bar{x}_6) \wedge \underbrace{(\bar{x}_2 \vee x_3 \vee \bar{x}_7)}_{\text{Clause}}$$

literals

Can you set the variable $x_1, x_2 \ldots$ → {T, F} such that the formula comes out as T (True)

Satisfying assignment

NP: Guess whether $x_1 = T$ or F

guess $x_2 = T$ or F

⋮

Check formula → T → return YES

F → return NO

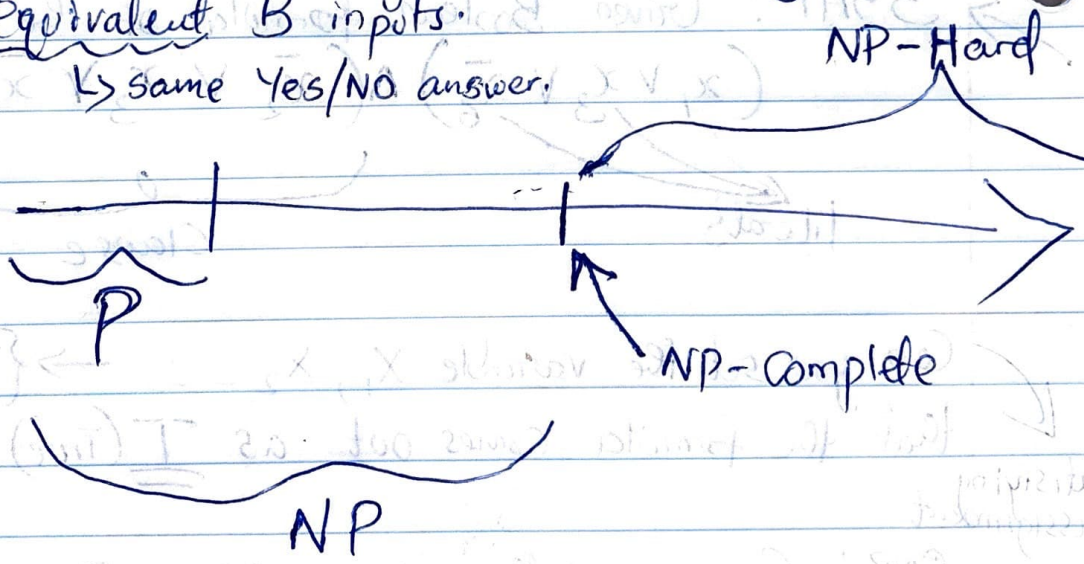
NP = {decision problems with poly-size certificates & poly-time verifiers for YES inputs}

→ X is <u>NP-complete</u> if $X \in NP$ & NP-hard.

→ X is <u>NP-hard</u> if every problem $Y \in NP$ <u>reduces</u> to X

→ <u>Reduction</u> from problem A → problem B = poly-time algorithm converting A inputs → <u>equivalent</u> B inputs.

↳ same Yes/NO answer.




⟹ $x \notin P$ unless $P = NP$

→ If $B \in P$ then $A \in P$

→ If $B \in NP$ then $A \in NP$

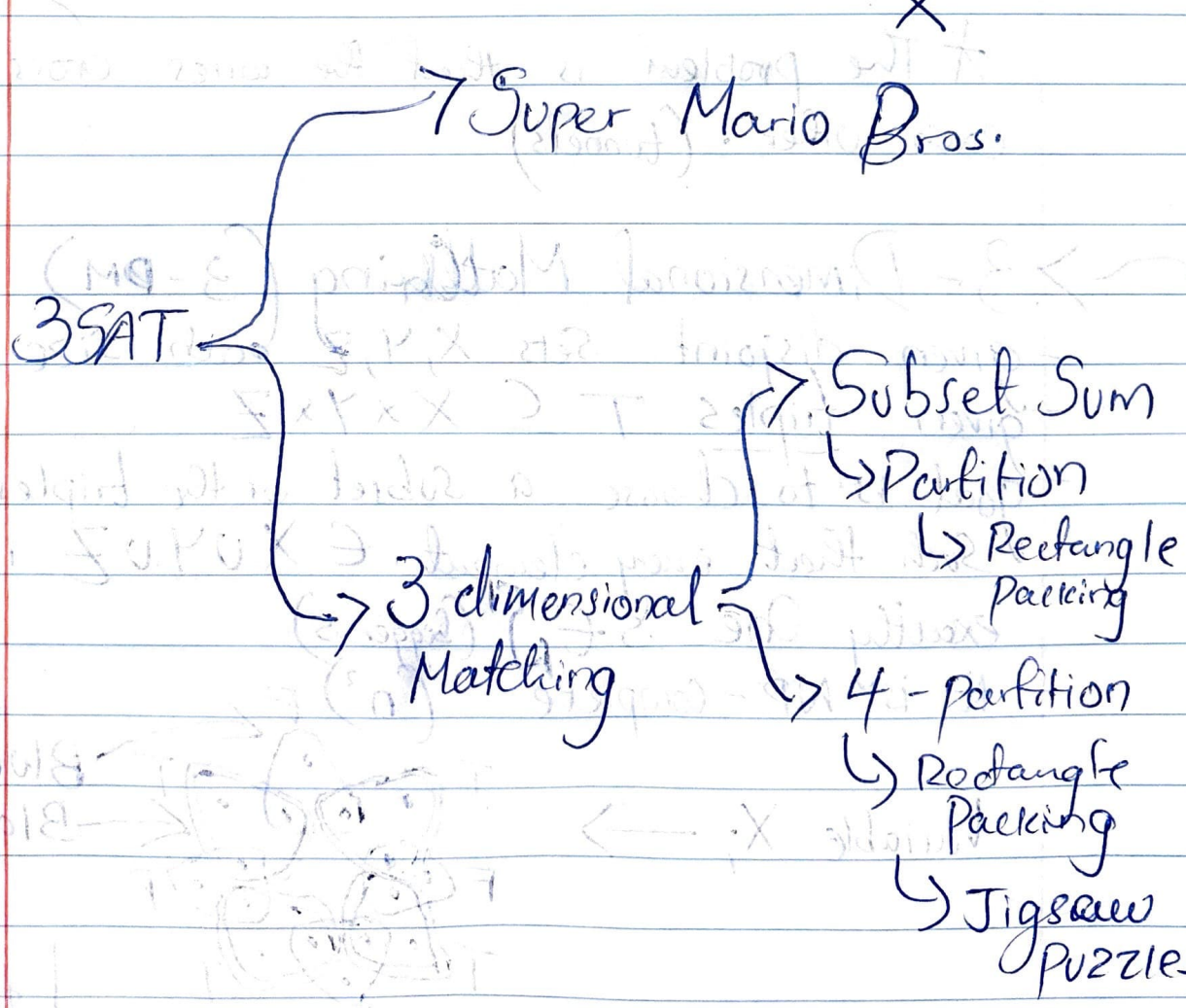
# Guide on how to prove:

How to prove X is NP-Complete:

1. Prove $x \in NP$ → non-det. algorithm
   → certificate + verifier
2. Prove $x \in NP$ hard

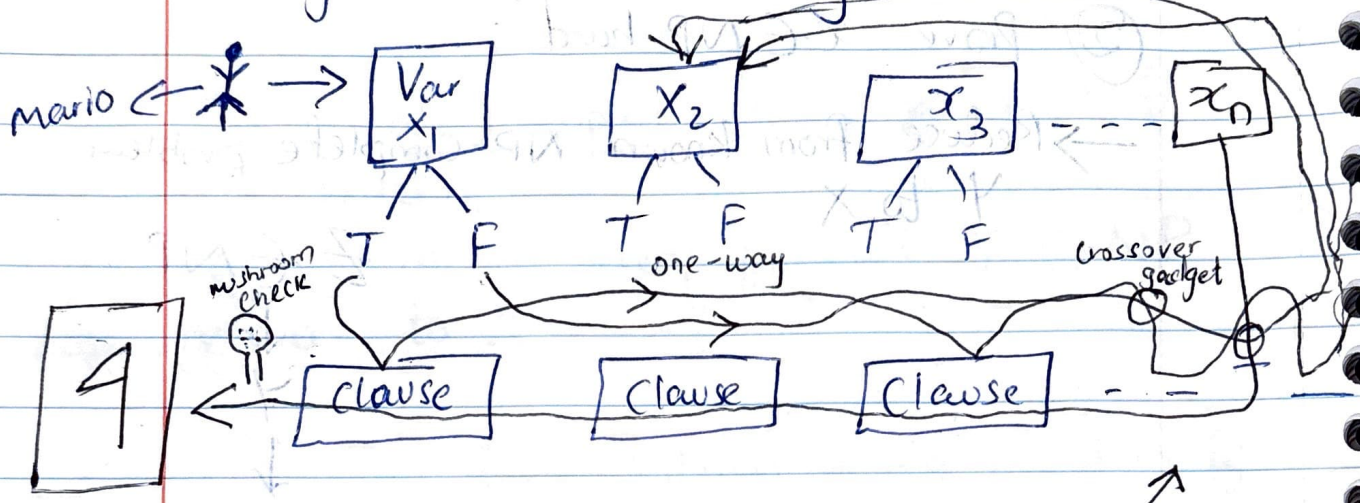
→ Reduce from known NP-Complete problem Y to X

$Z \in NP$
↓
Y
↓
X



## → Super Mario Bros : is NP-hard

- generalize to arbitrary screen size ($n \times m$)




* The problem is that the wires cross each other. (tunnels)

## → 3-Dimensional Matching (3-DM)

- given disjoint sets $X, Y, Z$ each size $n$
- given triples $T \subseteq X \times Y \times Z$
- goal is to choose a subset of the triples $S \subseteq T$ such that every element $\in X \cup Y \cup Z$ is in exactly one $s \in S$ (bigger $S$)
- It is NP-complete ($n^3$)

Variable $X_i$ →

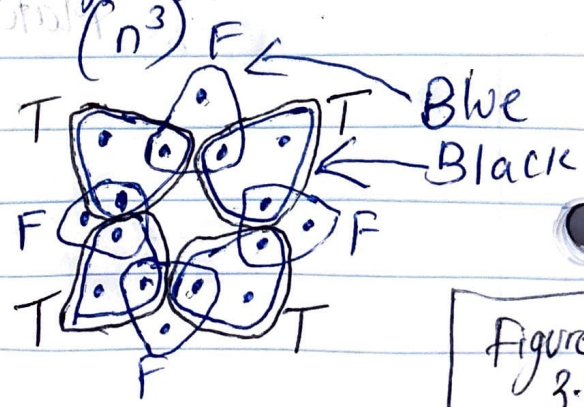


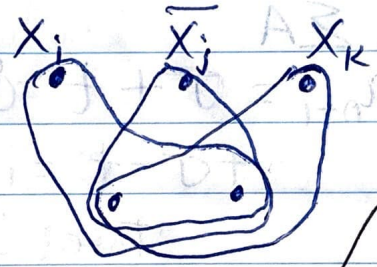

Figure 3.1

let's say Black - True
Blue - False.

$2 \underline{n} X_i$

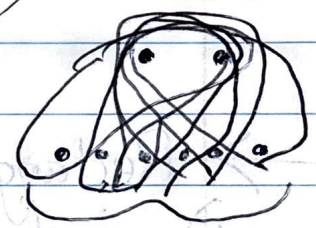
↳ # ocurrences of $X_i$ in the formula (clauses)

→ Clause $X_i \vee \overline{X_j} \vee X_K$




from figure 3.1

Garbage Collection



all T & F dots

$X \sum_{X} n_X$ - No. of clauses.

→ <u>Subset sum</u>:

given $n$ integers $A = \{a_1, a_2 \ldots a_n\}$ and target sum $t$ is there a subset $S \subseteq A$ such that

$$\sum_1 S = \sum_{a_i \in S} a_i = t$$

- Weakly NP-hard

- View numbers in base $b = 1 + \max n_{X_i}$

Reduction: - triple$(x_i, x_j, x_k) \rightarrow$ 0000|000|00|000

$(i, j, k)$

$t = 1111111111$

$= \sum_i b_i$ $\qquad = b^i + b^j + b^k$

* Value of numbers is exponential that's why its weakly np-hard

→ Partition Problem : given $A = \{a_1, a_2 \ldots a_n\}$, is there a subset $S \subseteq A$ such that

$$\sum S = \sum (A - S) = \frac{\sum A}{2}$$

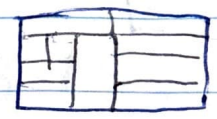
1) Reduction from Subset Sum.

- let $\sigma = \sum A$
- add $a_{n+1} = \sigma + t$ & add $a_{n+2} = 2\sigma - t$

$+\sigma - t$ $\qquad +t$

Total $= \sigma$

→ Rectangle packing : $R_i$ →



1) reduction by partition

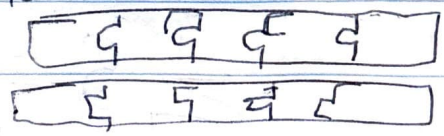
Weakly NP-hard

$a_i \rightarrow$ 1 $3a_i$

$T = 2$ $3t = \frac{3}{2} \sum A$

→ Jigsaw Puzzles partition does not work

NP-complete



~~Strong~~ Strongly NP-hard

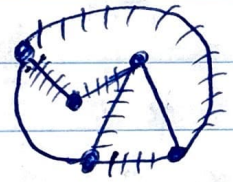
~~4 partion~~

4 partition – Given $n$ integers $A$, split $A$ into $n/4$ parts (quadruples) of same ~~size~~ sum

$$\frac{\sum A}{n/4} = t$$

Values $\leq n^{O(1)}$

$A \rightarrow$ Hamiltonian Cycle.

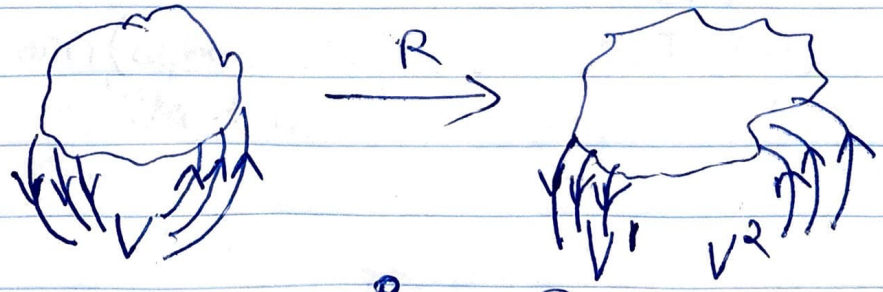


$B \rightarrow$ Hamiltonian Path.

$B \in NP$ $\qquad$ $B$ is hard ?

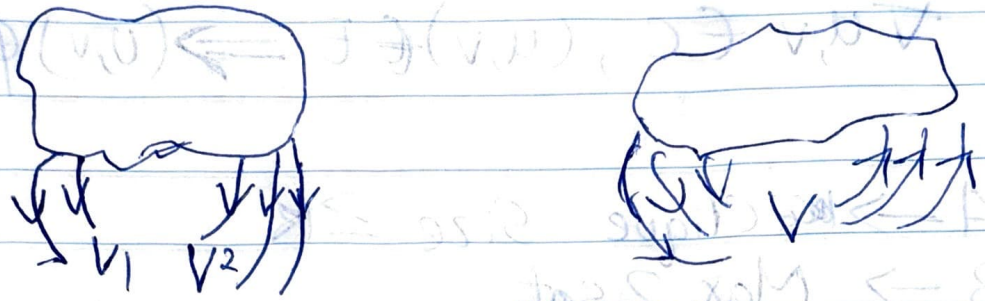
Input $A \rightarrow G = (V, E)$

$B \rightarrow G' = (V', E')$ Reduction




Cycle $\xrightarrow{R}$ Path

- the other way round

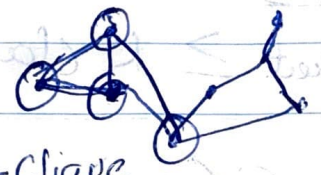



Path $\longrightarrow$ Cycle

$\therefore$ NP-Hard.

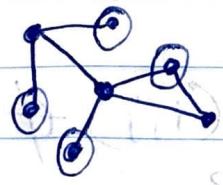
$\longrightarrow$ A $\rightarrow$ K-Clique? $G=(V,E)$

Clique —



0-clique
4-clique

B $\rightarrow$ Independent Set-K



— Independent set

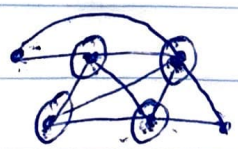
A — $G(V,E)$

$\downarrow$

B — $G'(V,\bar{E})$

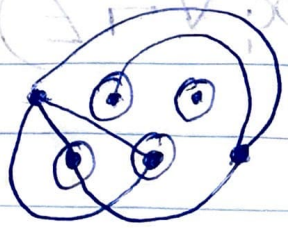
C — All edges Connected
I — No edges Connected

e.g



K-Clique

$\longrightarrow$



Independent Set.

Clique $C \rightarrow I$

$\forall u,v \in C, (u,v) \in E \Rightarrow (u,v) \notin E \; \forall u,v \in I$

$\rightarrow$ $A \rightarrow$ Clique Size $\geq K$

$B \rightarrow$ Max 2 SAT

$C = \{c_1, c_2, --- c_m\}$

$X = \{x_1, --- x_n\}$

$c_1 = (x_i \vee \neg x_j)$

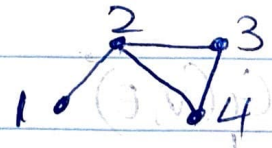
$\rightarrow$ $\exists$ assign such that $\geq K$ clauses satisfied

$\geq K$ - Clique $- G(V,E)$

$V' \subset V \quad |V'| \geq K$

$x_i \; \forall i \in V$, dummy literal $Z$

$> (\neg x_i \vee \neg x_j)$ such that $(i,j) \notin E$




$> (x_i \vee Z) \quad \forall i$

$> (x_i \vee \neg Z) \quad \forall i$

Clique $V' - |V'| = k$

$$x_i = \begin{cases} 1 & \text{if } i \in V' \\ 0 & \text{if } i \notin V' \end{cases}$$

$$z = 1$$

Clauses $- > (\neg x_i \vee \neg x_j) - |\bar{E}|$

$> (x_i \vee z) - |V|$

$> (x_i \vee \neg z) - |V'| = k$